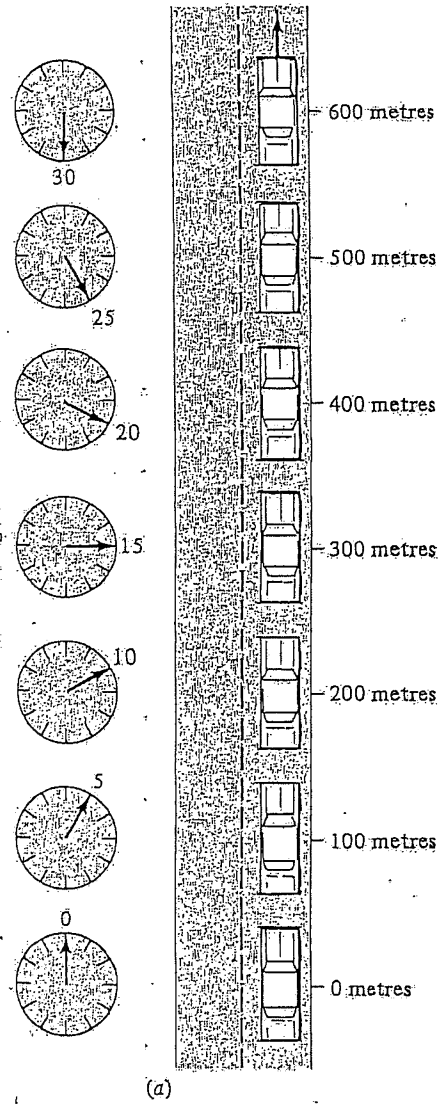
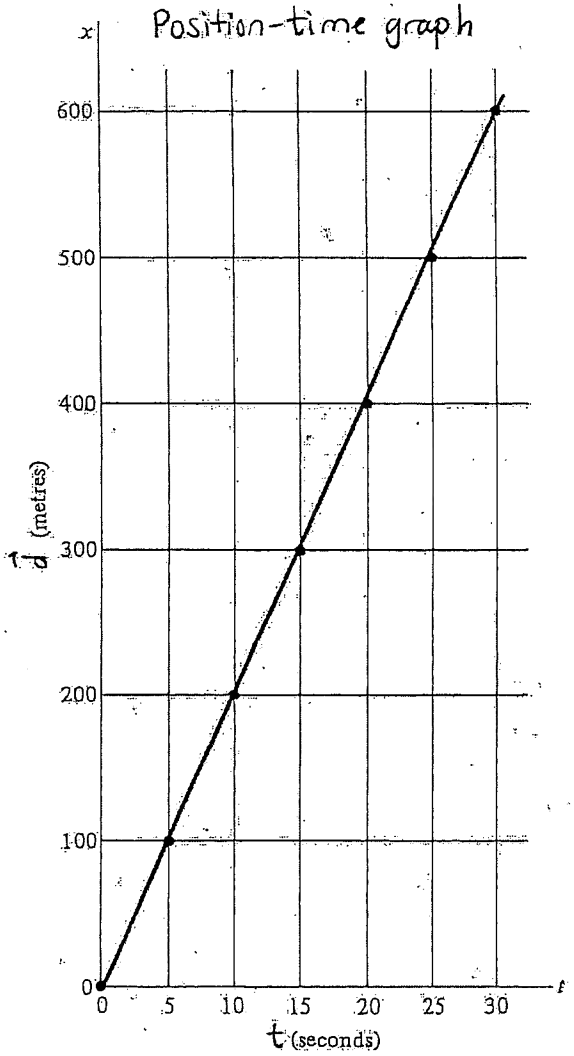
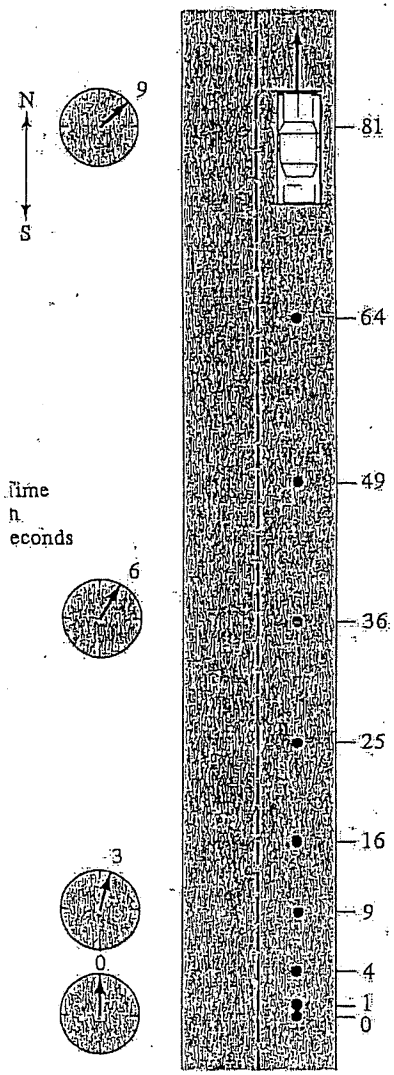

30
25
20
15
10
5
0
600 metres
500 metres
400 metres
300 metres
200 metres
100 metres
0 metres

(a)

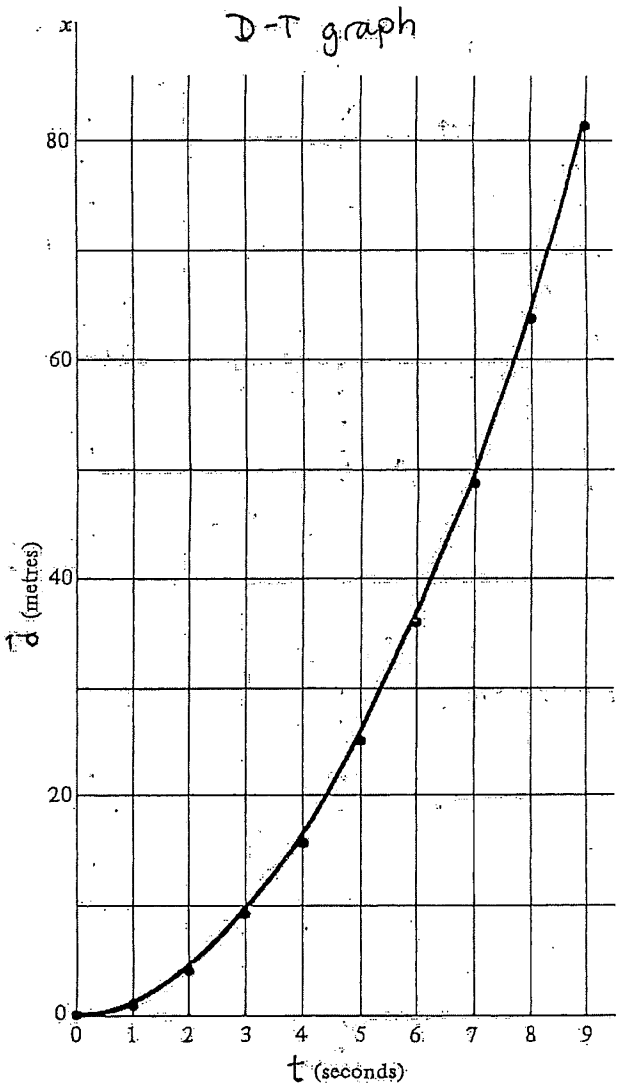

Position-time graph
x
600
500
400
300
200
100
0
d (metres)
0
5
10
15
20
25
30
t
t (seconds)

(b)


N
S
9
6
3
0
Time
in
seconds

81
64
49
36
25
16
9
4
1
0

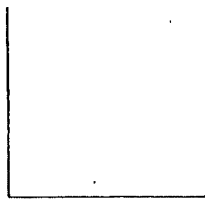
(a)


D-T graph
x
80
60
40
20
0
d (metres)
0
1
2
3
4
5
6
7
8
9
t
t (seconds)

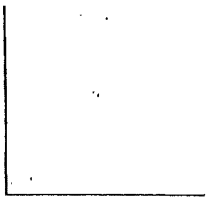
(b)

Position-Time Graphs when Velocity is NOT Constant

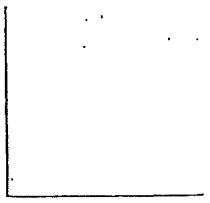
What does a position-time graph look like when the object is...



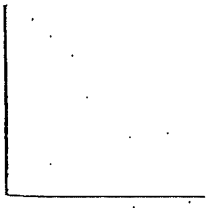
Going Forward, Speeding Up



Going Forward, Slowing Down



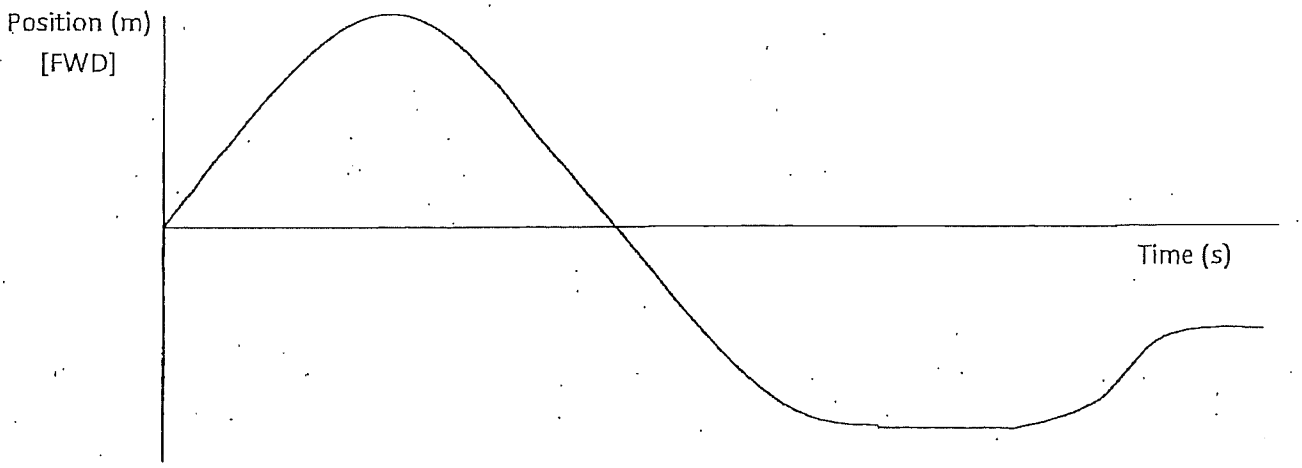
Going Backwards, Speeding Up



Going Backward, Slowing Down

Describe the motion represented in the position-time graph below. Identify where the object is:

- Moving backwards or forwards, or "at rest", and,
- Moving at a constant velocity, speeding up, or slowing down



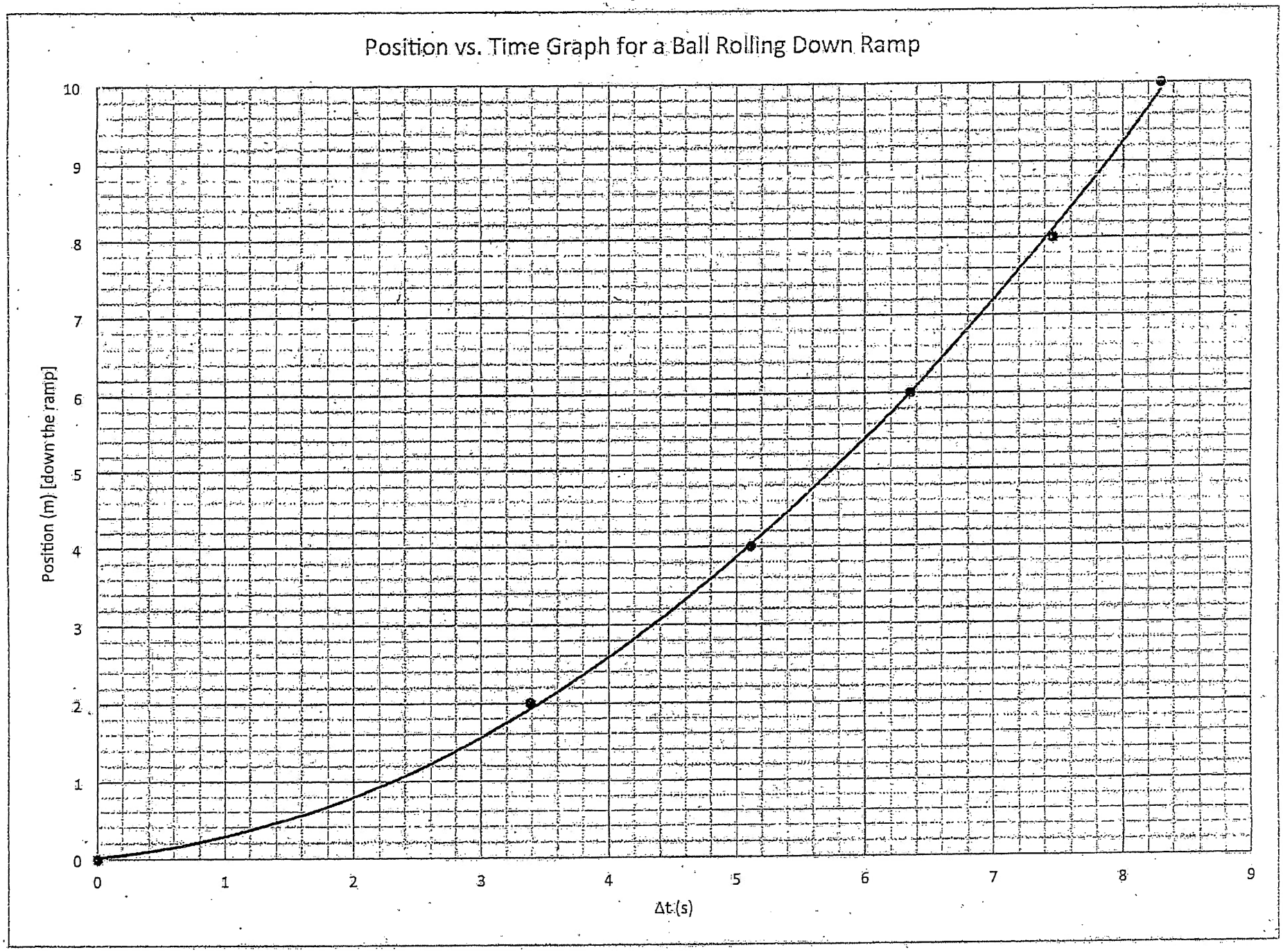
Kinematics Lab #2

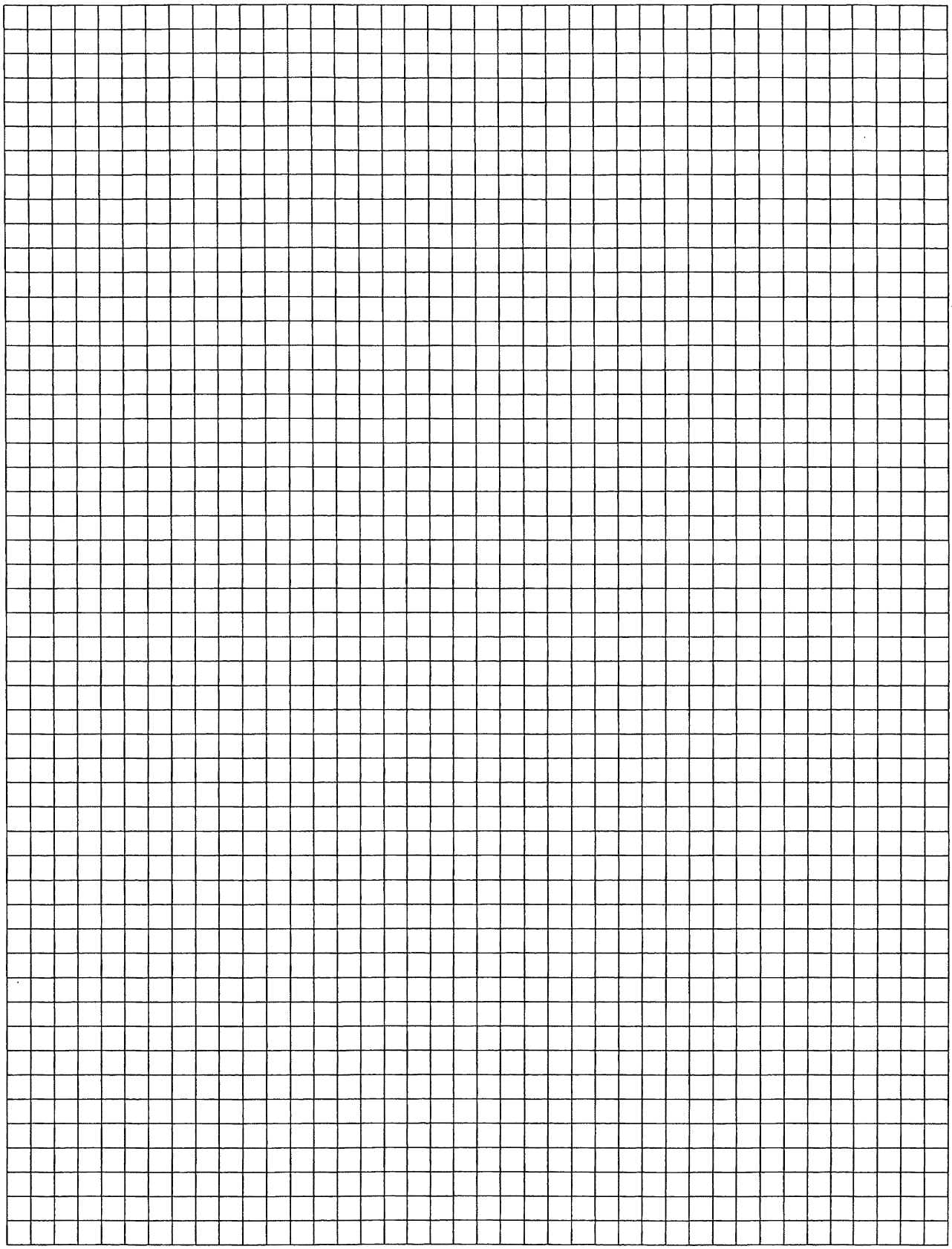
Sledding down a hill.

1. Based on your data table, plot the set of ordered pairs on a position vs. time graph (this is graph #1).
   NOTE: Make your graph take up the entire page – a larger graph leads to less error when drawing your tangent lines!
   In addition, it is easiest to draw your graph in "landscape": 
   rather than "portrait". 
2. Draw a curve of best fit for the points.
3. How can you tell, by looking at the curve, that the sled is speeding up, as it goes down the hill? Explain your answer, with reference to kinematics lab #1.
4. Draw at least 5 tangents to the graph. Try to make your tangent lines evenly spaced out. Make sure that the tangent lines are reasonably long.
5. Find the slope of each of the tangent lines. Include units. Show your work.
6. What does the slope of each of the tangent lines represent?
7. Make a new table of values:

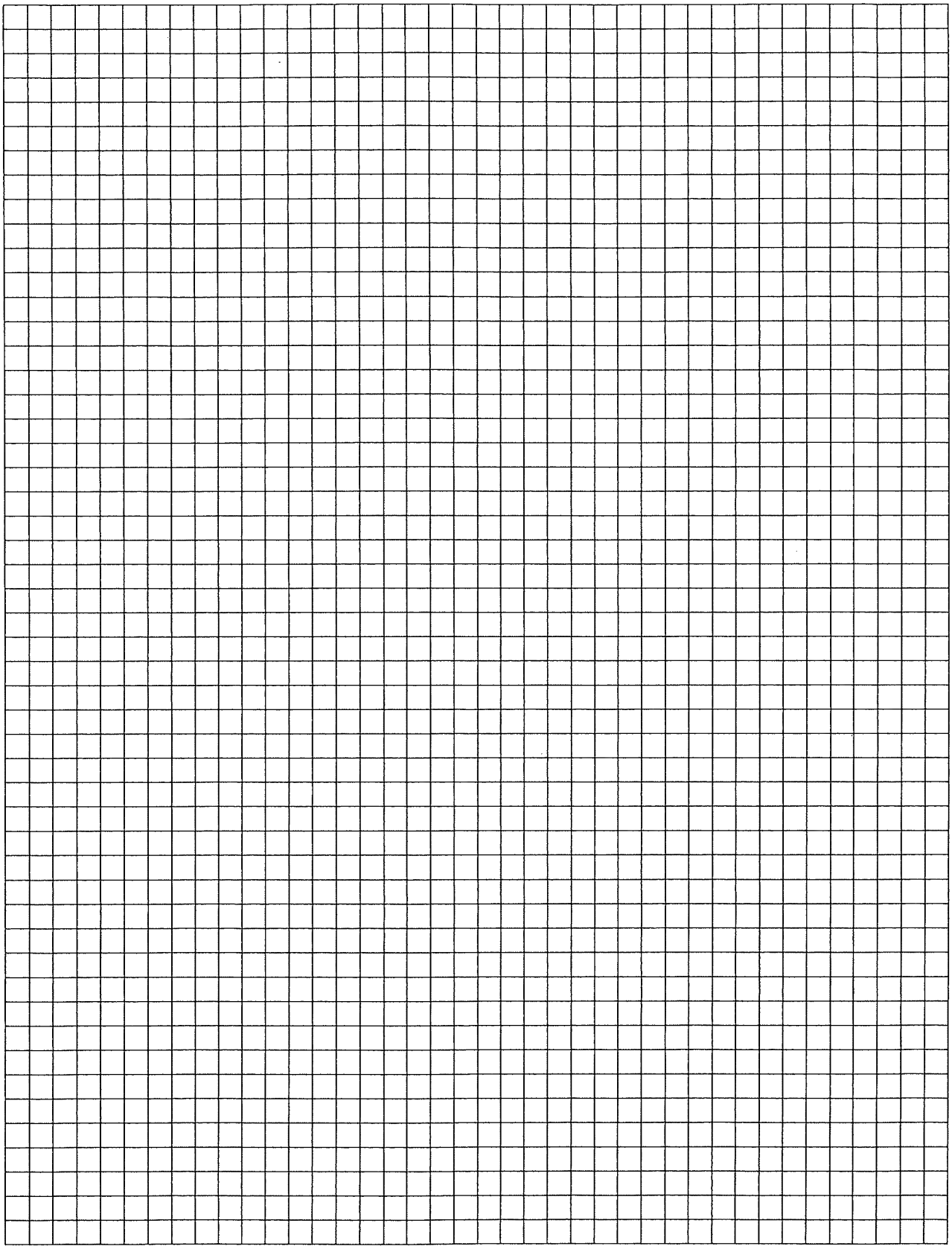
Time at which tangent touches the graph	Slope of tangent

8. Draw a new graph (graph #2), plot the ordered pairs from your table of values on this new graph.
9. Draw a line of best fit for this data set. Find the slope of the line of best fit. Include units. Show your work.
10. Draw a new graph (graph #3), plot the slope of the line of best fit from graph #2, at various times.
11. Draw a line through the points on your graph.
12. Give realistic sources of error for this lab. For each source of error, suggest plausible changes that could be made, to reduce this error.


Position vs. Time Graph for a Ball Rolling Down Ramp
Position (m) [down the ramp]
Δt (s)
0
1
2
3
4
5
6
7
8
9
10

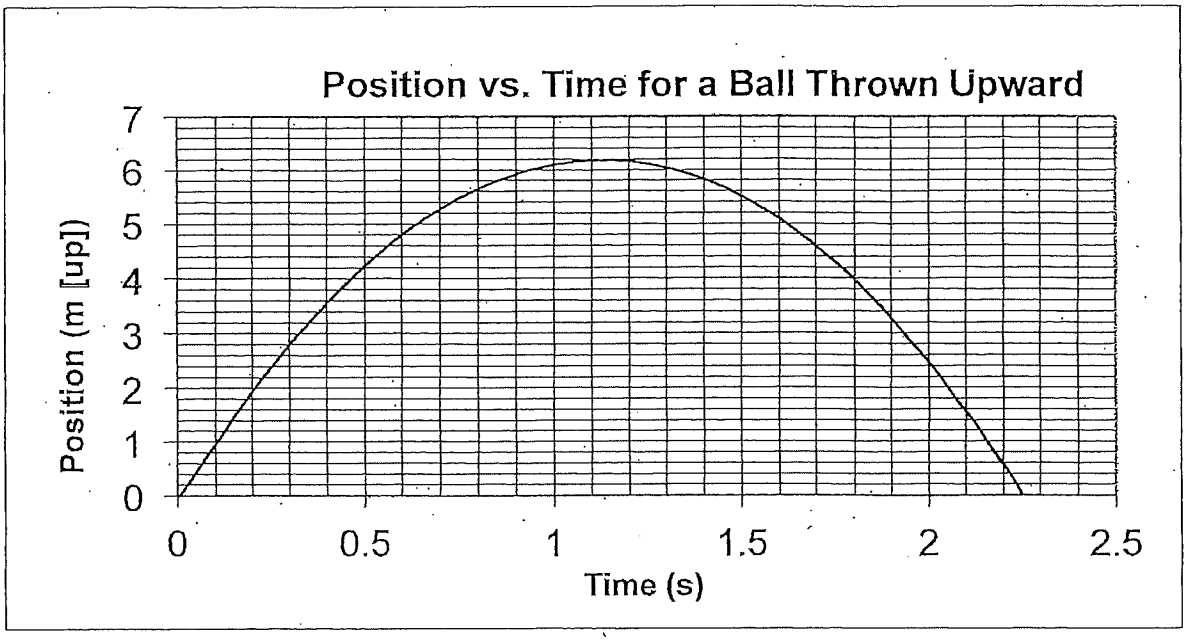


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Finding Velocity from a Curved Position-Time Graph



1. The ball is on its way up from 0s until _____ and it returns to the ground at _____. On the way up, its speed is _____ and you can tell because the slope is getting _____. On the way down, the speed is _____ and you can tell because the slope is getting _____. At the maximum height, its speed is _____.
2. Find the **average** velocity of the ball from 0s until 0.60 s.

3. Find the **average** velocity of the ball from 1.40 s until 2.2 s.

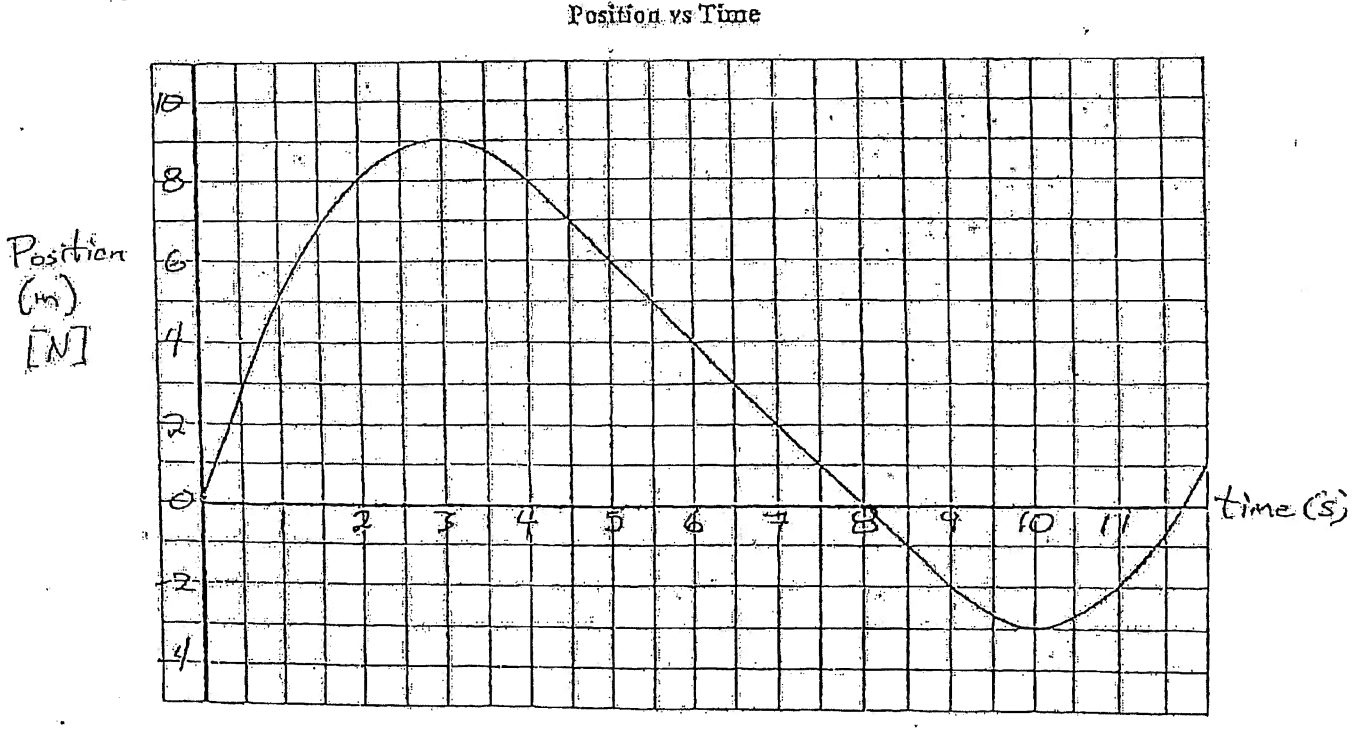
4. Find the **instantaneous velocity** of the ball at
   a. 0.10 s
   b. 1.60 s

Position vs. Time Graphs

For each of the given position vs. time graphs, answer the following questions:

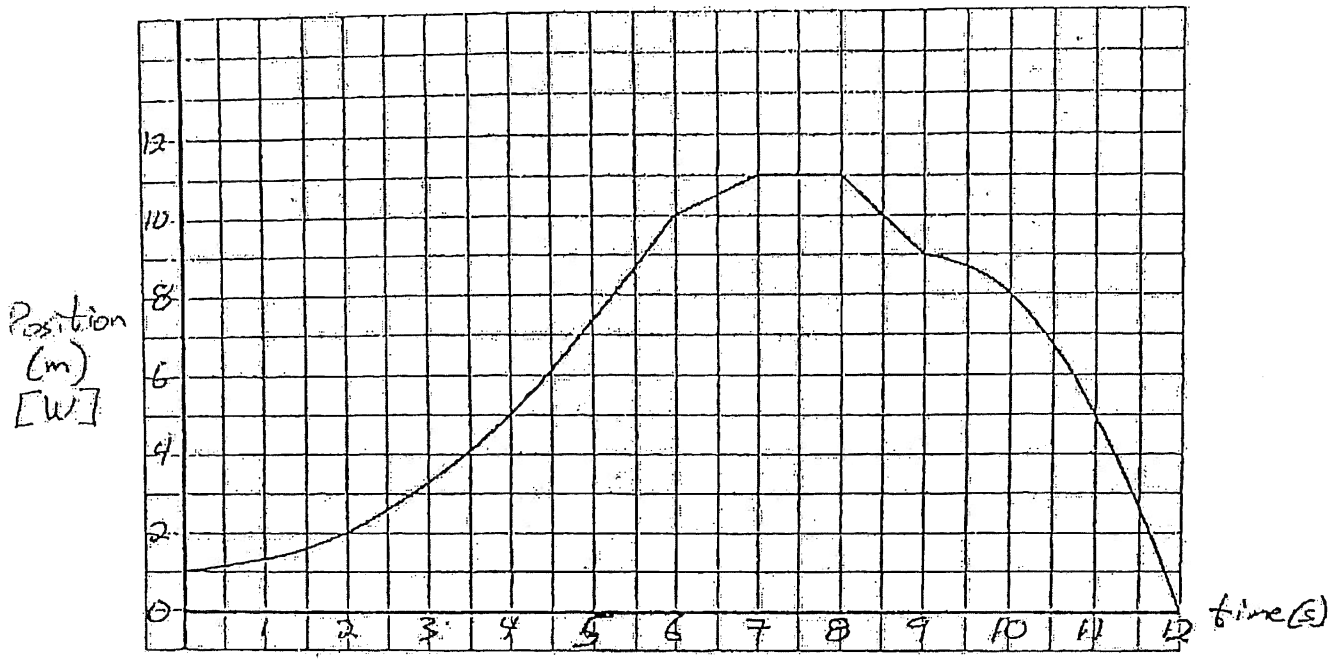
a) At what time(s) is the object stopped? (Or its velocity was 0)?
b) At what time(s) did the object return to its starting position?
c) What is the average velocity of the object during the following time intervals?
   i) 0 s to 3 s
   ii) 5 s to 7 s
d) What is the instantaneous velocity of the object at the following times?
   i) 3 s
   ii) 8 s
e) During what time interval(s) is the object moving in a positive direction? (you might think of this as moving forwards)
f) During what time interval(s) is the object moving in a negative direction? (you might think of this as moving backwards)
g) Write a paragraph, describing the motion of the object, indicating the direction of motion of the object and what type of motion is occuring (eg – the object is moving forwards and its speed is increasing)
h) Construct a velocity vs. time graph for the object's journey.

1.



2.

Position vs. Time



3. Answer the following general questions regarding Position vs. Time graphs:

a) How is it possible to tell by looking at a $\vec{d}-t$ graph when the object is stopped?

b) How is it possible to tell by looking at a $\vec{d}-t$ graph when the object is moving in a positive direction?

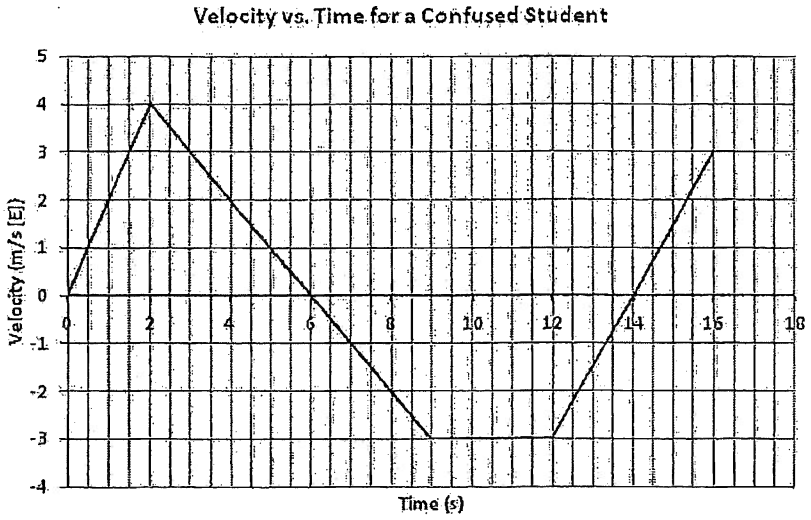
c) How is it possible to tell by looking at a $\vec{d}-t$ graph when the object is moving in a negative direction?

d) How is it possible to tell by looking at a $\vec{d}-t$ graph when the object speeding up?

e) How is it possible to tell by looking at a $\vec{d}-t$ graph when the object is slowing down?

Understanding Velocity vs. Time Graphs

Use the graph below to answer the following questions:



1. What direction is considered positive? ____ What direction must be negative? _____

2. When is the student travelling east?

_____

3. When are they travelling west?

_____

4. Describe the student's motion during section A (hint: look at the values for velocity): _____

5. Describe the student's motion during section B:

_____

6. What is the student's velocity at...

   6.0 s _____ 7.0 s _____ 8.0 s _____ 9.0 s _____

7. Describe the student's motion during section C:

_____

8. What is the student's velocity at...

   9.0 s _____ 10.0 s _____ 12.0 s _____

9. Describe the student's motion during section D: _____

10. What is the student's velocity at...

    12.0 s _____ 13.0 s _____ 14.0 s _____

11. Describe the student's motion during section E: _____

Conclusions:

If the object is speeding up, the line is moving _____ the x-axis

If the object is slowing down, the line is moving _____ the x-axis

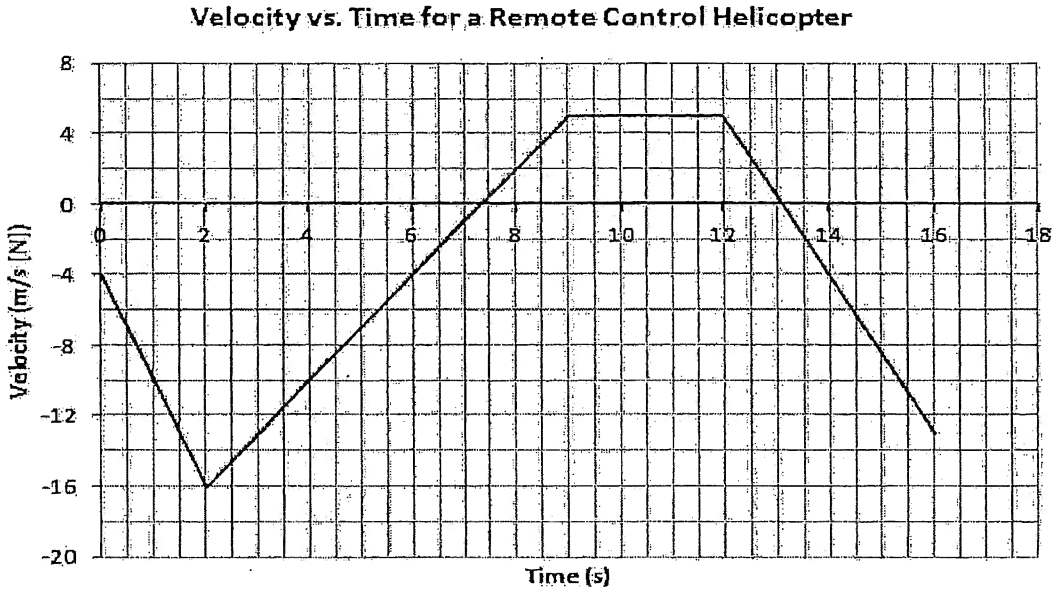
You can tell the object has turned around when the line _____ the x-axis

Uniform motion is shown by _____

You can tell the direction of travel by _____

The object is not moving when _____

Going Further



1. **Describe the motion of the helicopter** as shown by this graph. Include the direction of motion, speeding up or slowing down or constant speed, and direction of acceleration (which is not necessarily the same direction as velocity).

   Example: For t=0->2s: going S, increasing velocity, acceleration is south

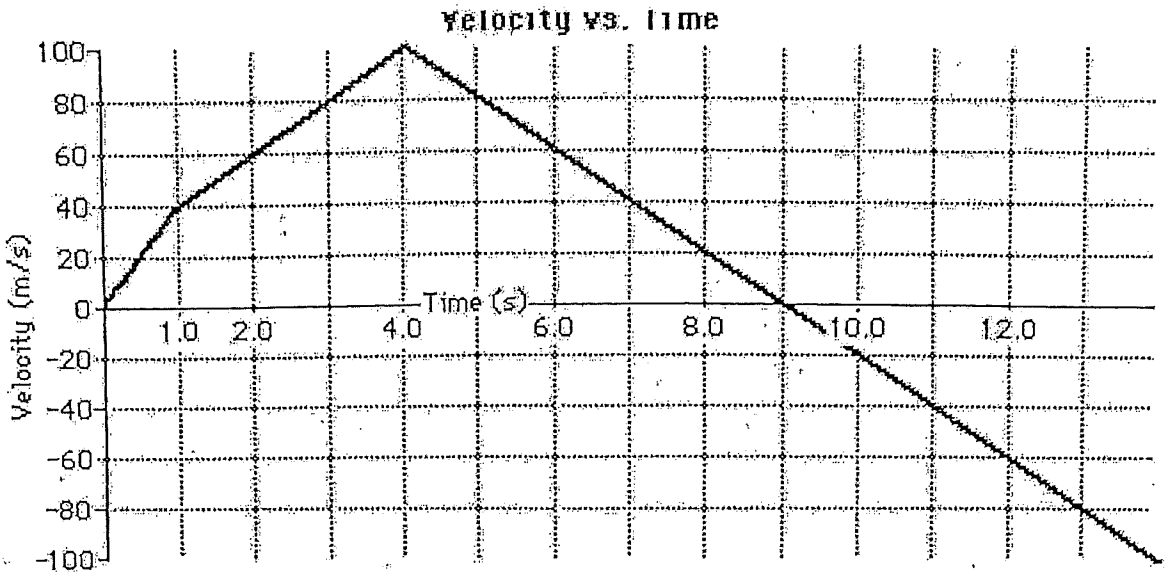
2. **Determine the acceleration** of the helicopter for the following segments:
   a. 0.0 – 2.0 s
   b. 2.0 s – 6.0 s
   c. 8.0 s – 9.0 s
   d. 12.0 s – 13 s
   e. 13.0 s – 16.0 s

3. Hopefully you found that your answers for b and c were the same. Does this mean that the helicopter was moving in the same way during both intervals? Explain.

4. Similarly, your answers for d and e were the same. Explain why, with reference to the helicopter's motion.

Interpreting Velocity-Time Graphs

The motion of a two-stage rocket is portrayed by the following velocity-time graph.



Several students analyze the graph and make the following statements. Indicate whether the statements are correct or incorrect. Justify your answers by referring to specific features about the graph.

Student Statement	Correct? Yes or No

1. After 4 seconds, the rocket is moving in the negative direction (i.e., down).

Justification:

2. The rocket is traveling with a greater speed during the time interval from 0 to 1 second than the time interval from 1 to 4 seconds.

Justification:

3. The rocket changes its direction after the fourth second.

Justification:

4. During the time interval from 4 to 9 seconds, the rocket is moving in the positive direction (up) and slowing down.

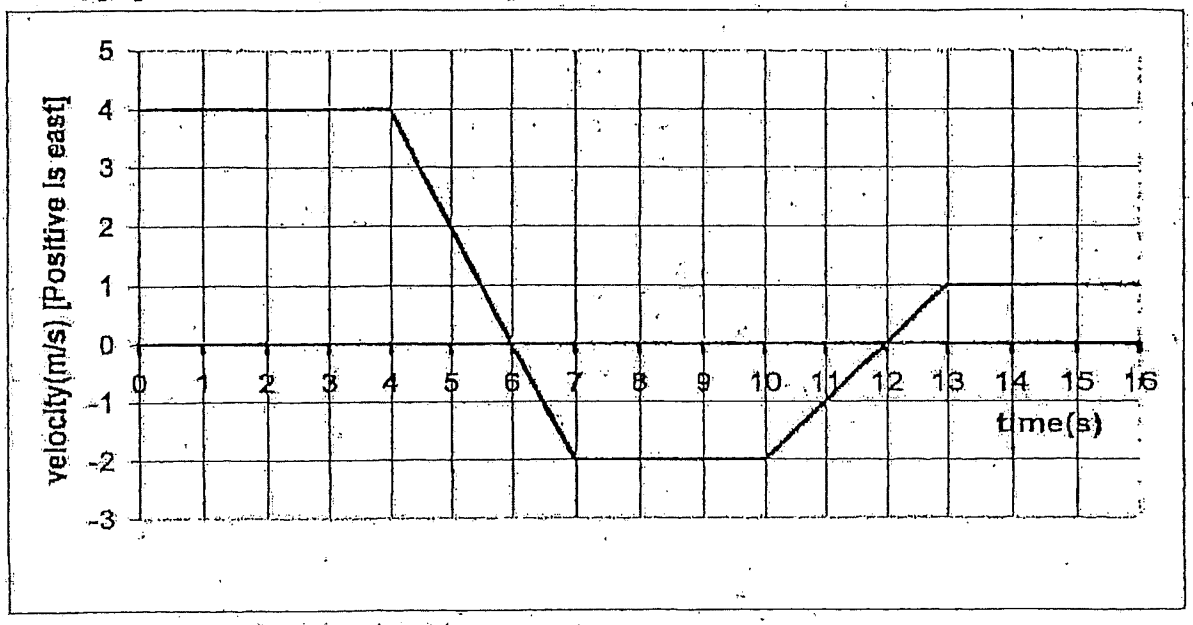
Justification:

5. At nine seconds, the rocket has returned to its initial starting position.

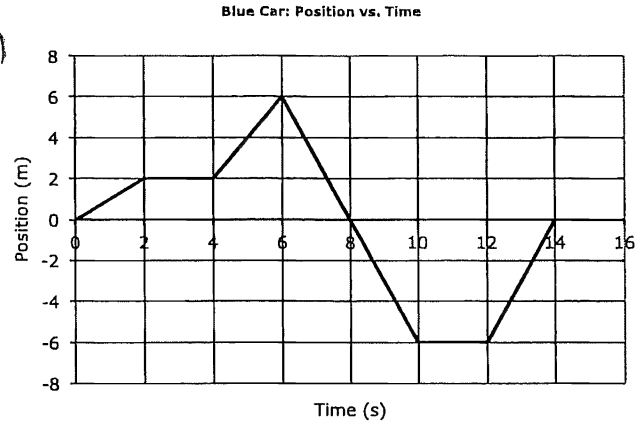
Justification:

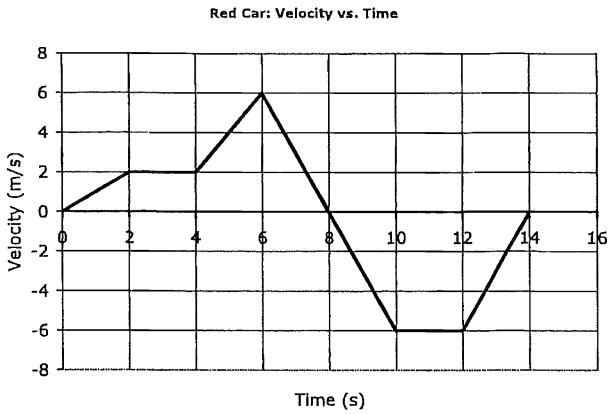
Velocity-time Graphs

Use the graph below to answer all of the questions in full detail.



1. At what time(s) is the object at rest?
2. At what time(s) is the object travelling east?
3. At what time(s) is the object travelling west?
4. What is the acceleration at 6.0 seconds?
5. What is the average acceleration for the entire 16 seconds?
6. What is the displacement for the first 6.0 seconds
7. What is the displacement for the first 10 seconds?
8. What is the velocity at 11 seconds?
9. What is the average velocity for the entire 16 seconds?
10. At what time(s) is the velocity of the object the largest
11. At what time(s) is the acceleration a positive value east?
12. At what time(s) is the acceleration a positive value west?





The two graphs above may look similar, but they actually represent two very different motion patterns. Calculate the answers to the following for both the Red car and the Blue car:

	BLUE CAR	RED CAR
velocity at t = 3 seconds		
average velocity from t = 0 s to t = 2 s		
velocity at t = 5 seconds		
average velocity from t = 6 s to t = 10 s		
maximum speed for the whole trip		
In which interval(s) was the car's velocity negative		
displacement during the first 2 seconds		
displacement during the first 4 seconds		
During which time interval(s) was the car's velocity zero		
During which intervals was the car's acceleration zero		

SPH3U: Graphing Review

This exercise will help you put together many of the ideas we have come across in studying graphical representations of motion. Before completing the table, examine the sample entries and be sure you understand them. Note that some information may be found on both graphs. Now fill in the missing entries in the table below – describe how you find the information.

How to get information from motion graphs

Information sought	Position-time graph	Velocity-time graph
Where the object is at a particular instant		
The object's velocity at an instant		
The object's acceleration (if constant)	Can't tell	Compute the slope
Whether the motion is uniform		
Whether the object is speeding up	Check whether the curve is getting steeper	
Whether the object is slowing down		
Whether the acceleration is constant		
The object's change in position		
The object's change in velocity		

Adapted from *Physics by Inquiry*, McDermott and PEG U. Wash, © John Wiley and Sons, *1996*

SPH3U: The Idea of Acceleration

Recorder: _____
Manager: _____
Speaker: _____
0 1 2 3 4 5

A: The Idea of Acceleration

Interpretations are powerful tools for making calculations. Please answer the following questions by **thinking and explaining** your reasoning to your group, rather than by plugging into equations. Consider the situation described below:

A car was traveling with a constant velocity 20 km/h sount. The driver presses the gas pedal and the car begins to speed up at a steady rate. The driver notices that it takes 3 seconds to speed up from 20 km/h to 50 km/h.

1. **Reason.** How fast is the car going 2 seconds after starting to speed up? Explain.

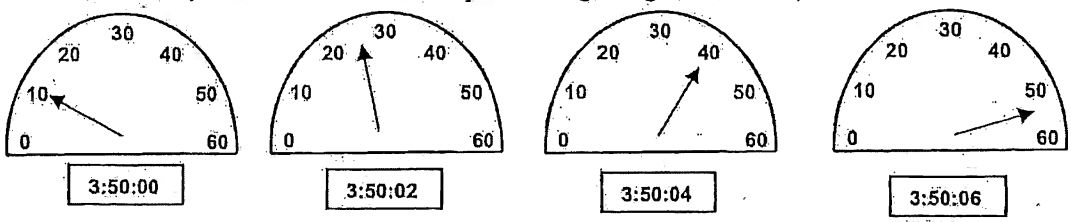
2. **Reason.** How much time does it take to go from 20 km/h to 75 km/h? Explain.

3. **Interpret.** A student who is studying this motion subtracts 50 – 20, obtaining 30. How would you interpret the number 30? What are its units?

4. **Interpret.** Next, the student divides 30 by 3 to get 10. How would you interpret the number 10? (Warning: don't use the word *acceleration*, instead explain what the 10 describes a change in. What are the units?)

B: Watch Your Speed!

Shown below are a series of images of a speedometer in a car showing speeds in km/h. Along with each is a clock showing the time (hh:mm:ss). Use these to answer the questions regarding the car's motion.



1. **Reason.** What type of velocity (or speed) is shown on a speedometer – average or instantaneous? Explain.

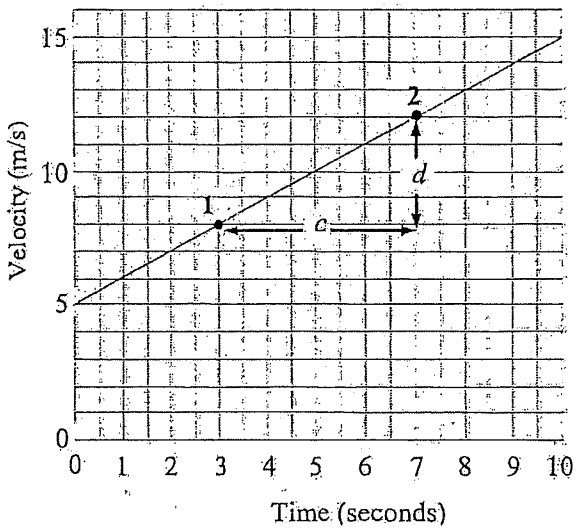
2. **Explain.** Is the car speeding up or slowing down? Is the change in speed steady?

3. **Explain and Calculate.** Explain how you could find the acceleration of the car. Calculate this value and write the units as (km/h)/s.

4. **Interpret.** Albert exclaims, "In our previous result, why are there two different time units: hours and seconds? This is strange!" Explain to Albert the significance of the hours unit and the seconds unit. The brackets provide a hint.

C: Interpreting Velocity Graphs

To the right is the velocity versus time graph for a particular object. Two moments, 1 and 2, are indicated on the graph.



1. **Interpret.** What does the graph tell us about the object at moments 1 and 2?

2. **Interpret.** Give an interpretation of the interval labelled c. What symbol should be used to represent this?

3. **Interpret.** Give an interpretation of the interval labelled d. What symbol should be used to represent this?

4. **Interpret.** Give an interpretation of the ratio d/c. How is this related to our discussion in part A?

5. **Calculate.** Calculate the ratio d/c including units. Write the units in a similar way to question B#3.

6. **Explain.** Use your grade 8 knowledge of fractions to explain how the units of (m/s)/s are simplified.

$$(m/s)/s = \frac{m}{s} \div s = \frac{m}{s} \div \frac{s}{1} = \frac{m}{s} \times \frac{1}{s} = m/s^2$$

17

The Direction of the Acceleration Vector

The general rule is:

If an object is **slowing** down, then its acceleration is in the **opposite** direction of its motion (velocity).

When an object is **speeding** up, the acceleration is in the **same** direction as the motion (velocity).

Consider the data tables below. Sketch A and B on one graph. Sketch C and D on another graph.

Example A

Time (s)	Velocity (m/s)
0	0
1	2
2	4
3	6
4	8

Example B

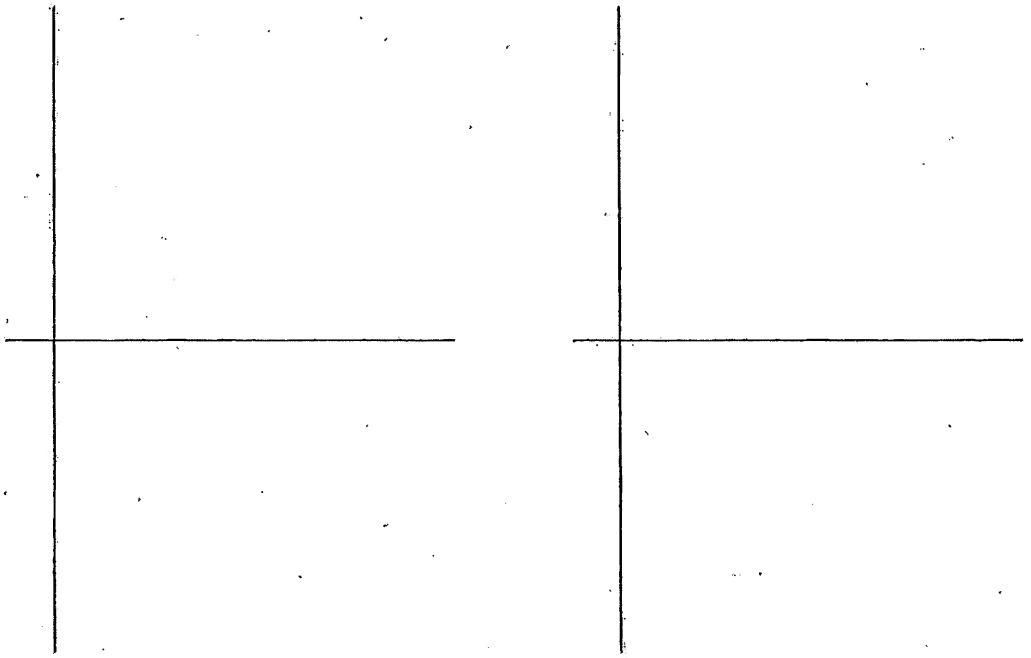
Time (s)	Velocity (m/s)
0	-8
1	-6
2	-4
3	-2
4	0

Example C

Time (s)	Velocity (m/s)
0	8
1	6
2	4
3	2
4	0

Example D

Time (s)	Velocity (m/s)
0	0
1	-2
2	-4
3	-6
4	-8



Can you explain why A and B are both positive acceleration, and why C and D are both negative acceleration?

Direction of acceleration:	SPEEDING UP	SLOWING DOWN
Moving in positive direction (+ v)	Positive acceleration	Negative acceleration
Moving in negative direction (- v)	Negative acceleration	Positive acceleration

Pierre Elliott Trudeau High School

Acceleration Problems

Answer the practice questions on a separate page. For each problem be sure to include a sketch to illustrate the situation and the information given in the question.

Sample Problems:

1. Getting onto the highway, a van accelerates from 60 km/h to 110 km/h in 30 seconds. What is its acceleration? (Express your answer in km/h/s)

2. Christine Nesbitt (winner of Olympic gold in 2010) starts at 1.5 m/s [FWD] and then accelerates at an average rate of 2.5 m/s^2 [FWD] for 1.5 s. What is her final velocity?

Practice Problems:

3. In a race, a car travelling at 100 km/h comes to a stop in 0.00139 hours. What is the average acceleration?

4. A track athlete at the end of her run goes from 12 m/s to a stop in 3.5 s. What is their average acceleration?

5. A car getting onto the highway accelerates at 15 km/h/s. How long will it take them to go from 40.0 km/h to 112 km/h?

6. A skydiver pulls their parachute, causing an acceleration of 2.5 m/s^2 [up]. Five (5.0) seconds later he is clocked at 6.5 m/s [down]. What was his speed when he pulled the cord?

7. A robot in a competition will be going at a constant speed and then when they get to a specific spot, they will accelerate. It needs to hit a top speed of 95 km/h in 20.0 s in order to win. If its original constant speed is 1.0 m/s and its maximum acceleration is 1.2 m/s^2, will it succeed? If not, what initial speed must it have?

8. The velocity of a roller coaster is measured at 25 m/s [up] on its way up a hill. Six (6.0) seconds later it is clocked on its way down the hill at 14 m/s [down]. What is the average acceleration of the roller coaster?

9. Young enjoys break dancing. In order to practice, he occasionally does back flips moving at 2 m/s [N] and then at 1.5 m/s [S] within a 5 s period. What is his acceleration?

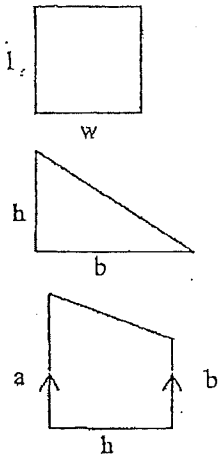
10. Two friends were riding a gazelle quickly through the African savannah. If they were traveling at 22 m/s [W] and then began to accelerate at 1.3 m/s^2 [E], what would their final velocity be after 32 s?

11. Suddenly, a jaguar appears out of nowhere traveling at 14 m/s [S]. If it accelerates at 1.6 m/s^2 [N], what would its velocity be after 14 s?

Analyzing Velocity vs. Time Graphs

Useful Knowledge:

Useful Formulas:

Name of Shape	Shape	Formula for Area of Shape
		

emember:

-
-
-

One other useful piece of knowledge:

Acceleration is

)w, given the velocity vs. time graph on the following page, generate a placement vs. time graph and an acceleration vs. time graph.

Understanding transition graphs DT - VT - AT

What does a horizontal line mean...

On a DT graph

What does a horizontal line mean...

On a VT graph

What does a horizontal line mean...

On an AT graph

What does a positive slope mean on a DT graph ?

What does a negative slope mean on a DT graph ?

What does a positive slope mean on a VT graph ?

What does a negative slope mean on a VT graph ?

How does an AT graph look if acceleration was not constant?

What does a curve mean on a DT graph?

Velocity is increasing in the positive direction.

VT graph looks like: DT graph looks like:

Velocity is increasing in the negative direction.

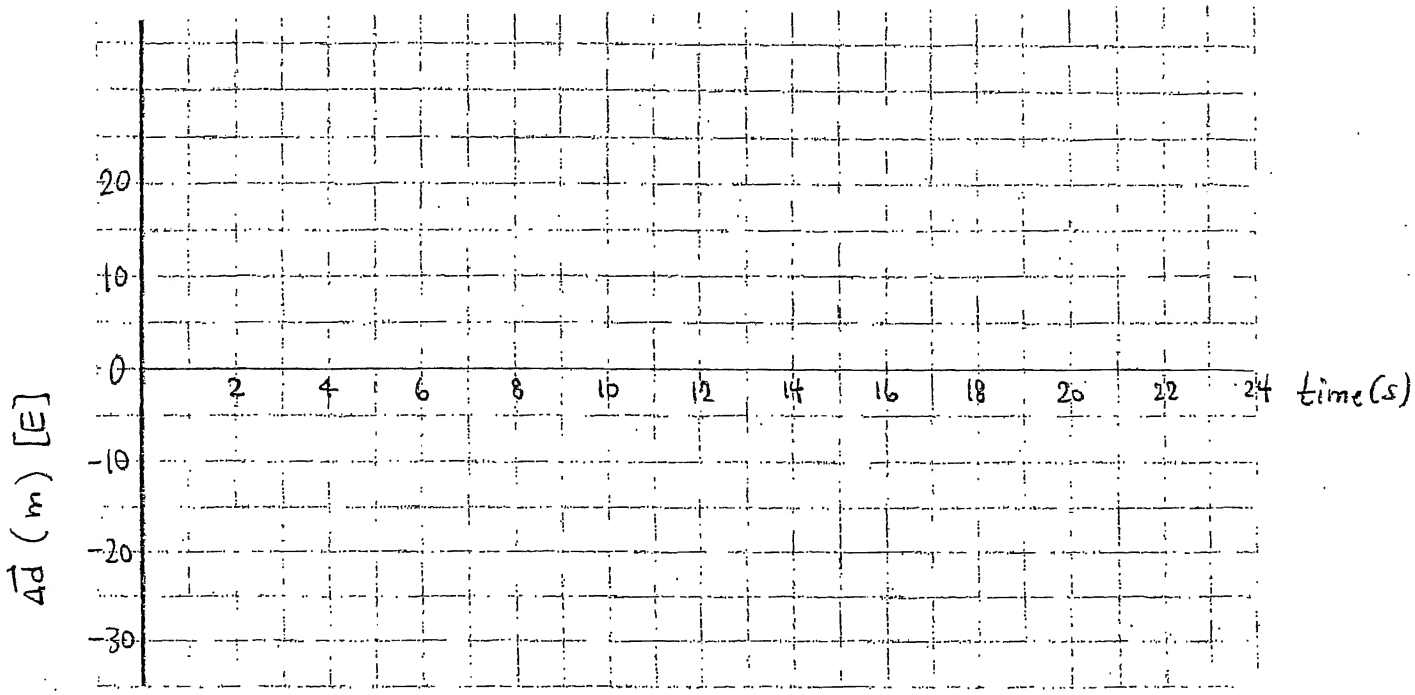
VT graph looks like: DT graph looks like:

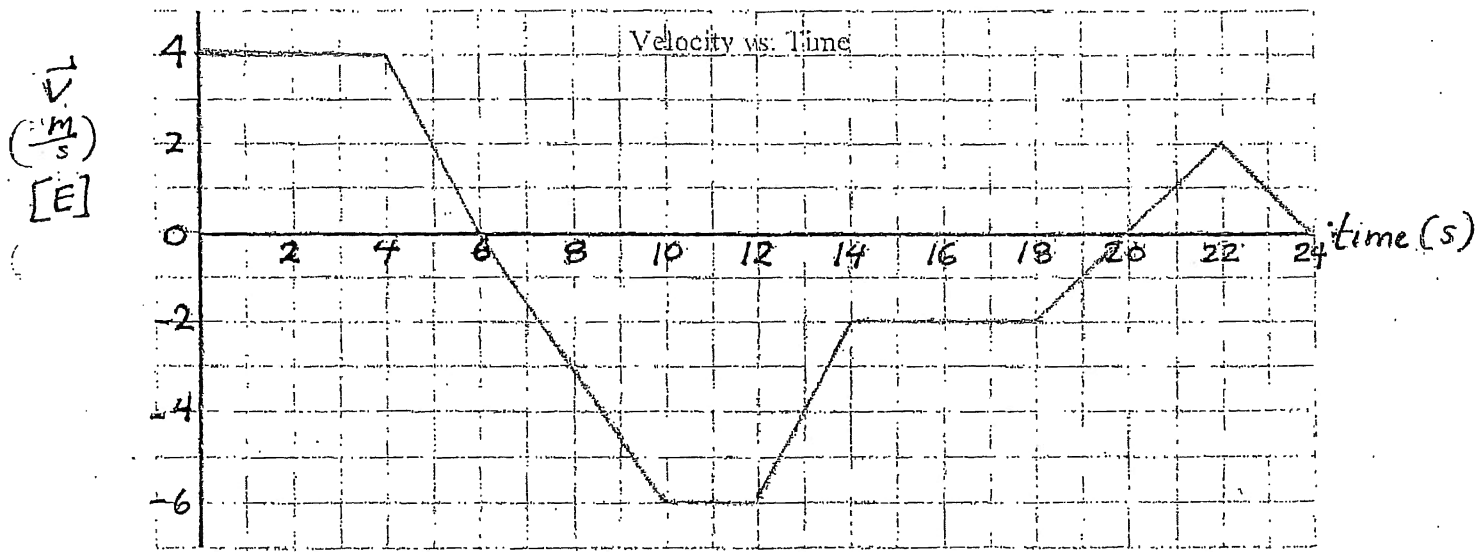
Velocity is decreasing in the positive direction.

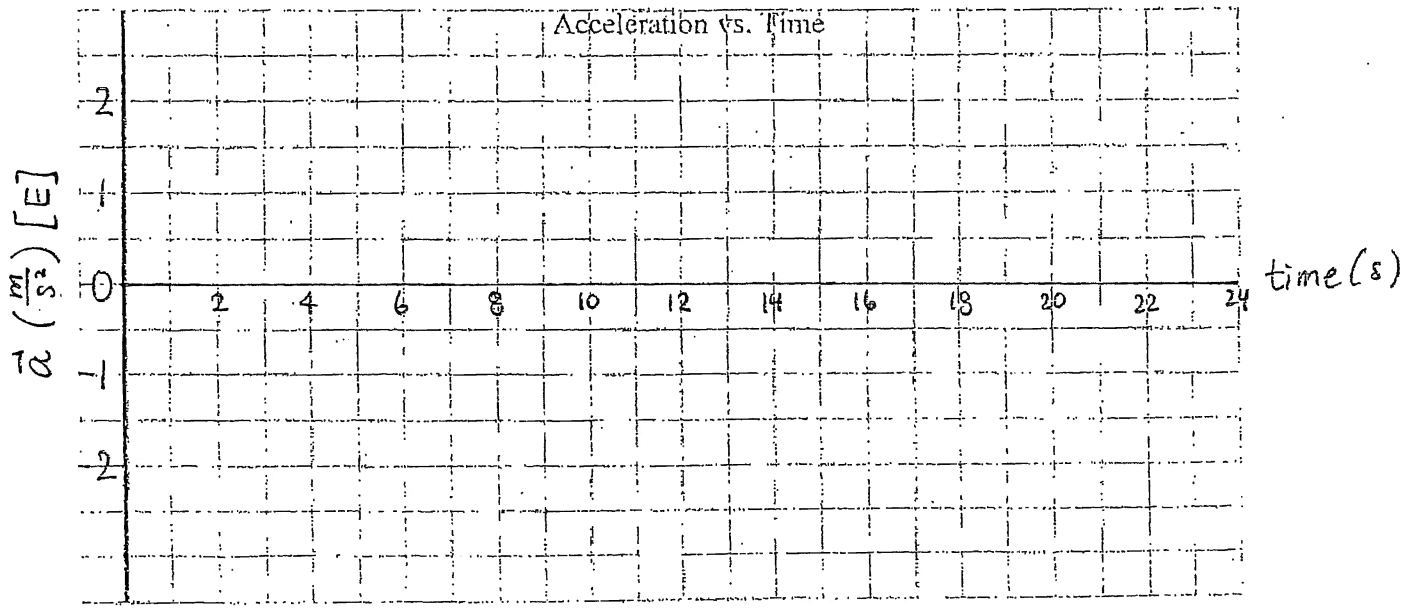
VT graph looks like: DT graph looks like:

Velocity is decreasing in the negative direction.

VT graph looks like: DT graph looks like:

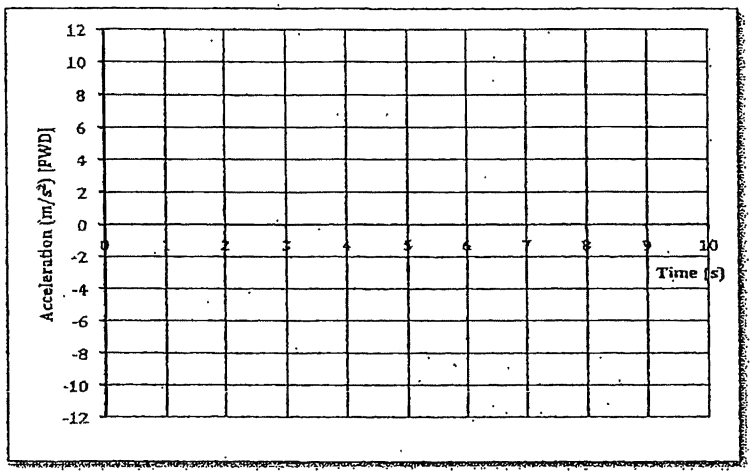

20
10
0
-10
-20
-30
2
4
6
8
10
12
14
16
18
20
22
24
time (s)
Δd (m) [E]

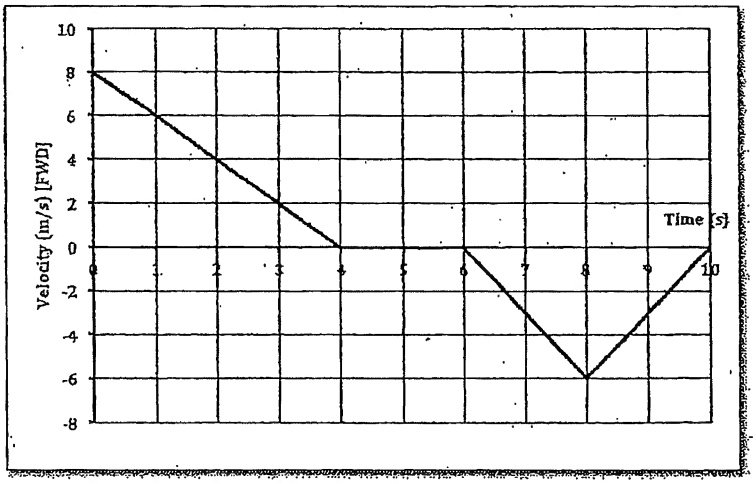

Velocity vs. Time
4
2
0
-2
-4
-6
2
4
6
8
10
12
14
16
18
20
22
24
time (s)
v (m/s) [E]

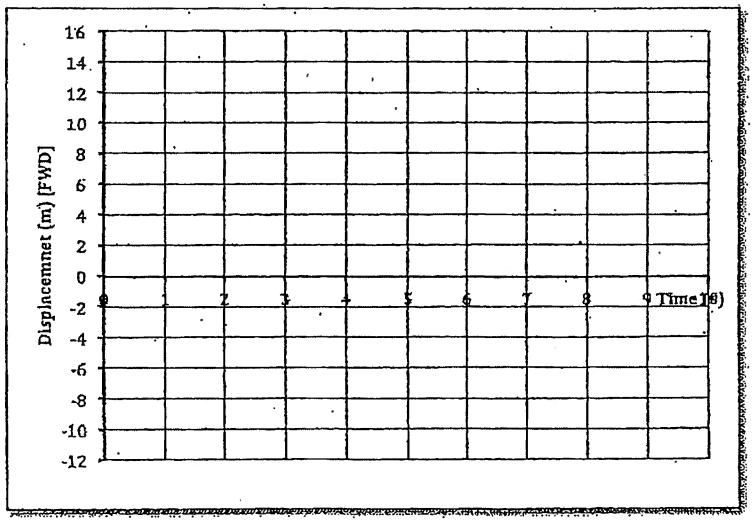

Acceleration vs. Time
2
1
0
-1
-2
2
4
6
8
10
12
14
16
18
20
22
24
time (s)
a (m/s²) [E]

TRANSITION GRAPHS

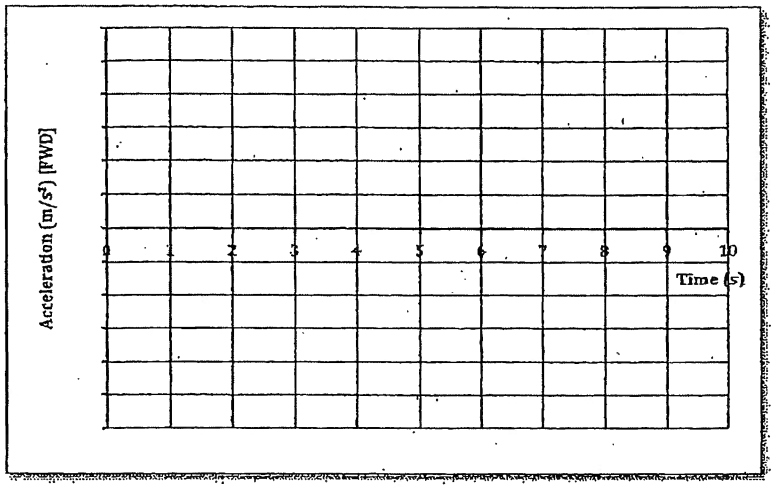
1. Draw the corresponding displacement - time graph and acceleration - time graph for the velocity-time graph shown. Show all calculations at side of graph for full marks.

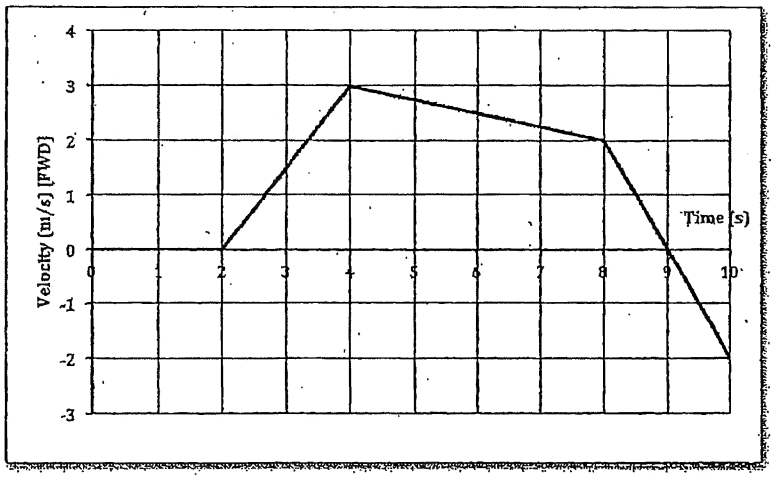


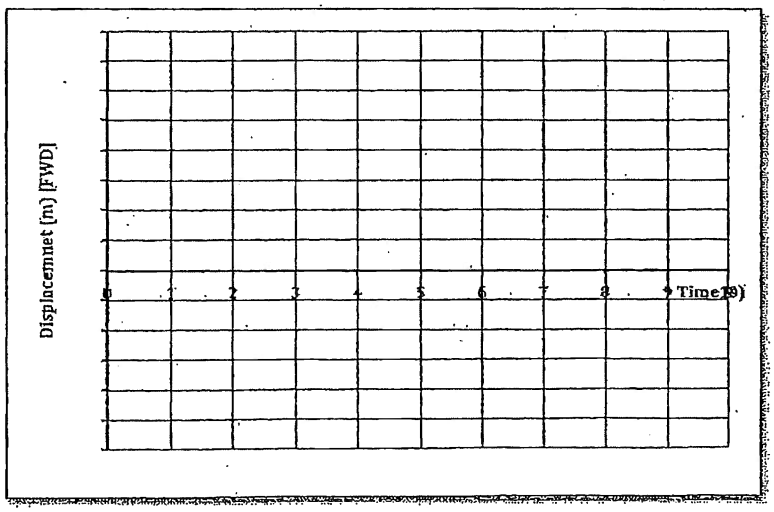




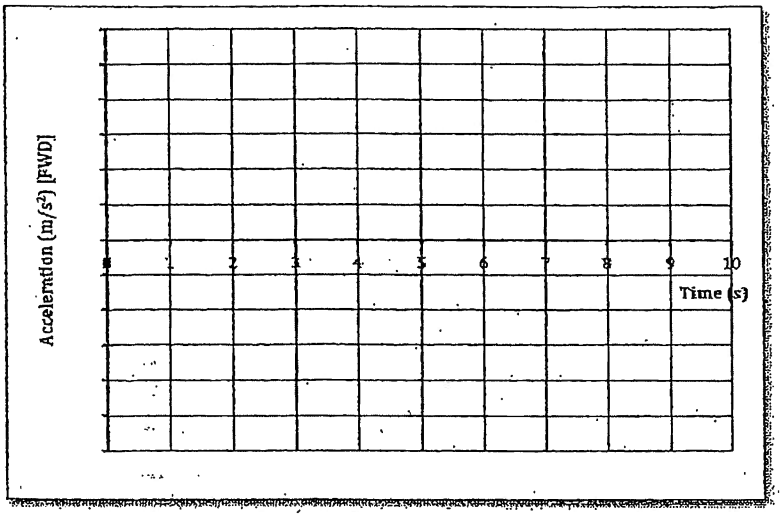
2. Draw the corresponding displacement - time graph and acceleration - time graph for the velocity-time graph shown. Show all calculations at side of graph for full marks.

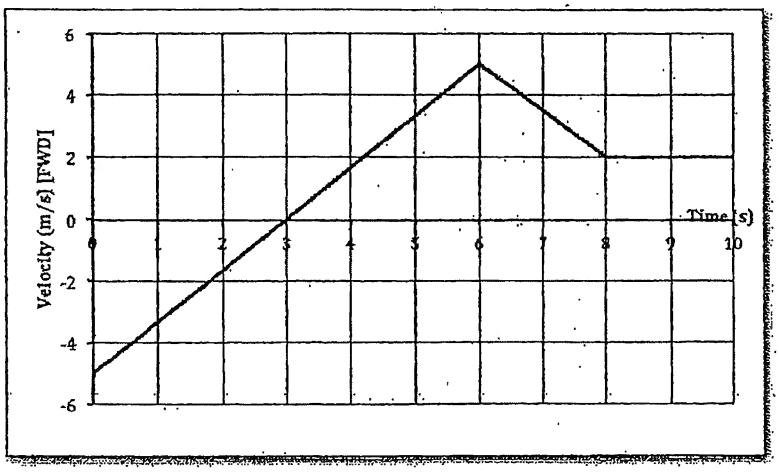


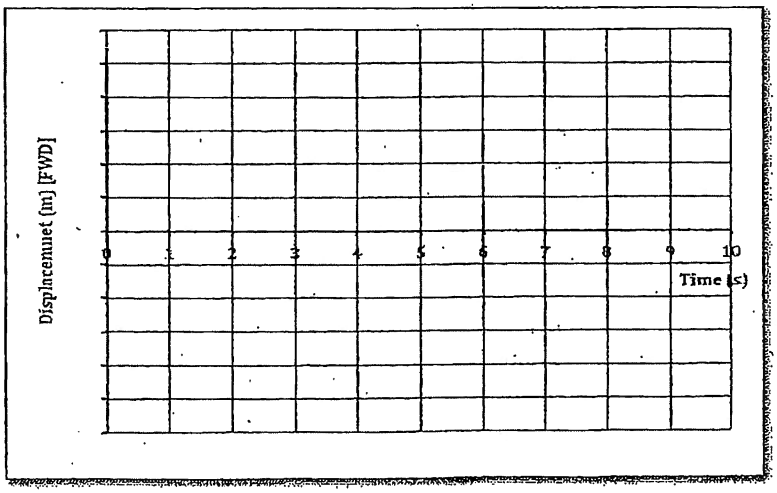




3. Draw the corresponding displacement - time graph and acceleration - time graph for the velocity-time graph shown. Show all calculations at side of graph for full marks.







TRANSITION GRAPHS

4. Draw the corresponding displacement - time graph and acceleration - time graph for the velocity-time graph shown. Show all calculations at side of graph for full marks.

